Sub Code: 23CS/IT/AD/AM1T04 **MIC23**

B. Tech I Semester Regular Examinations

INTRODUCTION TO PROGRAMMING

(CSE, AI&DS, AI&ML, IT)

(Model paper-I)

Time: 3 hours **Max Marks: 70M**

Note: 1. Question paper consists of two parts (**Part A and Part B**)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

		PART – A	**10 X 2 = 20 M**		
			CO	**BTL**	**Marks**
I.	**a)**	Write the differences between RAM and ROM	**CO1**	**L3**	**2 M**
	b)	What are the logical operators in C	**CO1**	**L1**	**2 M**
	c)	Explain go to statement with example	**CO2**	**L2**	**2 M**
	d)	Write a program to display whether a given year is leap year or not	**CO2**	**L3**	**2 M**
	e)	Explain strcmp() function.	**CO3**	**L2**	**2 M**
	f)	What is a subscript. How it is used to access an array element	**CO3**	**L1**	**2 M**
	g)	What is a preprocessor directive	**CO4**	**L1**	**2 M**
	h)	What is the purpose of free().	**CO4**	**L3**	**2 M**
	i)	Define bit field.	**CO5**	**L3**	**2 M**
	j)	What is the purpose of fseek()	**CO5**	**L1**	**2 M**
		PART – B	**5 X 10 = 50 M**		
		UNIT – I			
1.	**a)**	Explain the functional diagram of computer	**CO1**	**L2**	**5 M**
	b)	What are the characteristics of an algorithm? Write an algorithm for displaying the area of a triangle.	**CO1**	**L1**	**5 M**
		(OR)			
2.	**a)**	Explain the structure of C Program.	**CO1**	**L2**	**5 M**
	b)	Explain Bitwise operators. Write a program for demonstrating bitwise operators	**CO1**	**L2**	**5 M**
		UNIT – II			
3.	**a)**	Discus if else statement with an example program	**CO2**	**L6**	**5 M**
	b)	Explain switch statement. Write a program for displaying whether a given character is vowel or not using swith statement.	**CO2**	**L2**	**5 M**
		(OR)			
4.	**a)**	Differentiate while loop with do while loop.Demonstrate while and do while with an example	**CO2**	**L4**	**5 M**
	b)	Write a program to display whether a given number is prime or not.	**CO2**	**L3**	**5 M**
		UNIT – III			
5.	**a)**	What is an array? Explain the declaration, initialization of one dimenstional arrays.	**CO3**	**L1**	**5 M**
	b)	Write a program to multiply two two dimensional arrays.	**CO3**	**L3**	**5 M**
		(OR)			
6.	**a)**	Write a program for reversing a string with out using string handling function.	**CO3**	**L3**	**5 M**
	b)	Explain any five string handling functions defined in string.h library	**CO3**	**L2**	**5 M**
		UNIT – IV			
7.	**a)**	Explain the elements of user defined function.	**CO4**	**L2**	**5 M**
	b)	What is a recursion. Write a recursive program for Fibonacci series.	**CO4**	**L5**	**5 M**
		(OR)			
8.	**a)**	Discuss how arrays are represented using pointers	**CO4**	**L6**	**5 M**

	b)	Differenciate call by value and call by reference.	**CO4**	**L4**	**5 M**
		UNIT – V			
9.	**a)**	What is a structure? Explain the definition, initialization and accessing of structure variables	**CO5**	**L1**	**5 M**
	b)	Write a program for displaying division of a student using a structure containing regd no, name and marks in three subjects.	**CO5**	**L5**	**5 M**
		(OR)			
10.	**a)**	Explain functions used for input and output operations on fles	**CO5**	**L2**	**5 M**
	b)	Write a program for merging of two files	**CO5**	**L3**	**5 M**

***** End of the Question Paper *****

Sub Code: 23CS1T04,23IT1I04,23AD1T04,23AM1T04 **MIC23**

B. Tech I Semester Regular Examinations

INTRODUCTION TO PROGRAMMING

(CSE, AI&DS, AI&ML, IT)

(Model Paper-II)

Time: 3 hours **Max Marks: 70M**

Note: 1. Question paper consists of two parts (**Part A and Part B**)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

		PART – A			**10 X 2 = 20 M**
			CO	**BTL**	**Marks**
I.	**a)**	Write the differences between Compiler and Interpreter	**CO1**	**L3**	**2 M**
	b)	Explain conditional Operator	**CO1**	**L2**	**2 M**
	c)	Explain do while statement.	**CO2**	**L2**	**2 M**
	d)	Write a program to display whether a given number is even or odd	**CO2**	**L3**	**2 M**
	e)	What are the differences in reading of string using gets() and scanf() statement.	**CO3**	**L1**	**2 M**
	f)	What is an multidimensional array.	**CO3**	**L1**	**2 M**
	g)	What is a chain of pointers.	**CO4**	**L1**	**2 M**
	h)	Differentiate user defined and library functions.	**CO4**	**L4**	**2 M**
	i)	What is an array of structure	**CO5**	**L1**	**2 M**
	j)	What are the modes used in fopen().	**CO5**	**L1**	**2 M**
		PART – B			**5 X 10 = 50 M**
		UNIT – I			
1.	**a)**	What is a software? Explain the differences between application software and system software	**CO1**	**L1**	**5 M**
	b)	What are the symbols used in flowcharts. Draw the flow chart for displaying division of a student,	**CO1**	**L1**	**5 M**
		(OR)			
2.	**a)**	Explain the fundamental data types in C.	**CO1**	**L2**	**5 M**
	b)	Explain the Input and output statements in C.	**CO1**	**L2**	**5 M**
		UNIT – II			
3.	**a)**	Explain else if ladder. Write a program for displaying the roots of quadratic equation with else if ladder.	**CO2**	**L2**	**5 M**
	b)	Write a program for displaying the areas of different geometric shapes using switch statement.	**CO2**	**L3**	**5 M**
		(OR)			
4.	**a)**	Explain for statement. Write a program to generate n terms of the Fibonacci series.	**CO2**	**L5**	**5 M**
	b)	Write a program for displaying whether a given number is Armstrong or not.	**CO2**	**L3**	**5 M**
		UNIT – III			
5.	**a)**	Explain how the two dimensional arrays are declared, initialized and accessed.	**CO3**	**L2**	**5 M**
	b)	Write a program to sort an array.	**CO3**	**L3**	**5 M**
		(OR)			
6.	**a)**	What is a string? Explain the declaration, initialization, reading and displaying of a string.	**CO3**	**L1**	**5 M**
	b)	Write a program for sorting an array of strings.	**CO3**	**L3**	**5 M**
		UNIT – IV			
7.	**a)**	Discuss the types of functions based on input parameters and return values.	**CO4**	**L6**	**5 M**
	b)	Explain the storage classes.	**CO4**	**L2**	**5 M**
		(OR)			
8.	**a)**	What is a pointer ?Explain how the pointer is declared,	**CO4**	**L1**	**5 M**

		initialsed and accessing a variable through its pointer.			
	b)	Write a program for sorting an array using pointers.	**CO4**	**L3**	**5 M**
		UNIT – V			
9.	**a)**	Explain Nested Structures with an example.	**CO5**	**L2**	**5 M**
	b)	Differentiate structures and unions with an example program.	**CO5**	**L4**	**5 M**
		(OR)			
10.	**a)**	Write a program for copying contents of one file to another	**CO5**	**L3**	**5 M**
	b)	Explain the functions used for random access files.	**CO5**	**L2**	**5 M**

*** **End of the Question Paper** ***